

**Muhammad Sohail**  
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## SUMMARY

Full-stack Software Engineer with 7+ years of experience building scalable, cloud-native applications using Node.js, TypeScript, React/Angular, and SQL/NoSQL databases. Strong background in designing microservices, APIs, and performance-optimized systems deployed on AWS and Kubernetes, with hands-on experience in CI/CD, testing, and cross-functional collaboration. AI enthusiast exploring agent-based systems and Retrieval-Augmented Generation (RAG) to build practical, data-driven automation solutions, with a strong interest in impact-driven and sustainability-focused products.

## EDUCATION

### Università degli Studi di Firenze

*MS, Software, Science and Technology (In Progress)*

- Student Representative, Commissione Paritetica Docenti-Studenti (Joint Teacher-Student Committee) (*Feb 2026 – Present*) – serves as a liaison between the student body and faculty to monitor and improve teaching quality.

### Riphah International University

Spring 2018

*BS, Software Engineering (CGPA 3.05)*

## Certifications

AWS Certified Cloud Practitioner

MongoDB Certified

React Certification

Dart and Flutter: The Complete Developer's Guide

English Proficiency Test

AWS

MongoDB University

[TestDome](#)

Udemy

Duolingo

## AI & AUTOMATION

### SigenStor AI Agent: RAG-Driven Energy Consultant

- Engineered an AI agent using n8n and PostgreSQL to maintain session-aware state, leveraging historical dialogue to provide contextually accurate responses throughout the customer journey.
- Developed a RAG (Retrieval-Augmented Generation) pipeline using vector embeddings to ingest technical manuals and pricing data, enabling the LLM to deliver precise, zero-hallucination technical support.
- Integrated HomeWizard energy telemetry to automate residential system sizing, dynamically calculating battery capacity requirements based on 3-phase grid data and real-time load profiles.
- Programmed automated validation logic for installation constraints, including electrical cable cross-section requirements ( $5 \times 4\text{mm}^2$ ) and modular stack floor-loading limits (70kg per unit).

### Document Intelligence & Automated Form Processing

- Engineered an automated OCR pipeline using Google Cloud Vision, backed by an AWS Pub/Sub-based event pipeline, to digitize over 50k handwritten pamphlets from physical street campaigns, with throughput of up to 50,000 images per day, significantly reducing manual labor for administrative workflows.

- Integrated OpenAI models to perform entity extraction and KPI derivation from unstructured OCR text, enabling automated pre-filling of digital records.
- Created a dual-view interface where the system automatically pre-fills the digital form from the pamphlet image on one side and displays the actual image on the other, allowing a single operator to simply confirm the data and hit submit.

### Voice Intelligence & Real-Time Transcription

- Developed a microservice leveraging WebSockets to stream call recordings to Watson Speech-to-Text, enabling real-time transcription and automated compliance monitoring.
- Architected a security layer to identify sensitive information, such as unintended requests for passcodes or restricted data, within live audio streams.
- Optimized streaming performance for high-volume call environments by integrating REST API hooks with asynchronous WebSocket events.

## WORK EXPERIENCE

### Iterca Platform

Mar 2026 – Sep 2026

*Senior Software Engineer / Platform Engineer*

- **Cloud Infrastructure & CI/CD Automation:** Engineered a scalable deployment pipeline on AWS Amplify for a distributed Next.js architecture. Resolved complex serverless injection issues, ensuring the secure delivery of runtime configurations (Prisma, AWS SES, external APIs) to Node.js Lambdas.
- **Modern Web Architecture (SSR/RSC):** Architected the platform using TypeScript, React, and Next.js App Router. Seamlessly bridged React Server Components (SSR) with Client Components to optimize web performance, reduce payload sizes, and deliver a highly responsive user experience.
- **API Orchestration & Distributed Services:** Designed secure Node.js backend services to handle JWT authentication, magic-link provisioning, and third-party REST API orchestration (Calendly). Built robust data synchronization between decoupled frontend and backend applications.
- **Database Management & Caching Strategies:** Provisioned and managed AWS RDS PostgreSQL infrastructure, optimizing for high availability and strict cost controls within an AWS VPC. Implemented advanced Next.js caching strategies and custom invalidation rules to balance database load with real-time data accuracy.
- **Platform UX & Reusable Tooling:** Elevated platform usability by refactoring legacy server-side forms into reusable, state-driven React client components. Implemented comprehensive loading states, error boundaries, and seamless UI overlays to improve the end-to-end developer experience and user conversion flows.

### Trident Marketing

Sep 2018 – Present

*Tech Lead / Software Engineering Manager*

- Led a cross-functional engineering team delivering Angular and Node.js applications, coordinating backend, frontend, and QA development across multiple production systems.
- Architected scalable microservices and hybrid mobile solutions, leveraging Node.js, TypeScript, and MongoDB to support enterprise-level requirements. Designed system architecture patterns, API gateways, and authentication flows (JWT/OAuth) for secure, cloud-native applications.
- Integrated mobile apps with external booking systems and third-party services to streamline business operations (REST, WebSockets, ElasticSearch).

- Designed and co-architected the TRA API to manage real-time data flow between multiple apps and external booking systems, leveraging REST API principles for scalability and efficiency (Node.js, TypeScript).
- Designed the TRADB (MongoDB) schema for large-scale data handling and integrations.
- Optimized system performance and scalability by managing real-time synchronization across multiple Elastic indices, applying caching strategies, and tuning database queries.
- Enabled booking-to-sales conversion by updating household records, converting leads to members, and indexing data in the Manifest System.
- Built the Engage App using Angular and TypeScript, enabling agents to track customer data (e.g., demographics, call history) and manage bookings within the same platform, significantly streamlining campaign workflows and improving agent productivity.
- Engineered the Manifest System (Angular, Node.js) to track bookings, agent performance, commissions, and the overall sales lifecycle.
- Data Visualization: Highcharts, Chart.js, Recharts, dashboard analytics, interactive reporting, real-time data visualization.
- Developed the DNQ System, a multi-part survey platform:
  - Backend uses a tree data structure for complex survey logic.
  - Frontend built with Ionic (Angular, Node.js).
- Contributed to the Lead Input Application (LIA) UI for uploading leads to S3 and splitting large PDFs (Angular, Node.js).
- Created various survey management platforms using Ionic.
- Implemented a message broker for high-volume calls, integrating with TRADB for faster processing.
- Built WebSocket-based solutions for real-time notifications, integrated with REST APIs to enhance scalability (e.g., member reservations).
- *Quality Assurance:*
  - Engineered complex Data Merging Logic for Household/Lead entities, utilizing Mocha & Chai to validate 20+ edge cases (e.g., duplicate resolution, membership inheritance).
  - Implemented a robust Test-Driven Development (TDD) workflow using JSON fixtures to simulate API responses, ensuring zero-regression deployments.
  - Developed the logic for Dynamic Meeting Payloads, validating attendee filtering rules (primary vs. secondary contacts) through automated assertion suites.
  - *CI/CD & Reporting:* Integrated the test suite into Jenkins pipelines, configuring it to capture JSON test results and generate automated build reports for immediate feedback on code stability.
- *Python:*
  - Used ChatGPT for data analysis, KPI derivation, and pre-filling forms.
  - Developed a FastAPI-based rule engine to validate data extracted via OCR against historical records to determine user eligibility and system compatibility.
  - Wrote behavior-driven tests using Behave.
  - Performed data cleaning with Pandas and built Jupyter notebooks for analysis.
- *Infrastructure & CI/CD:*
  - Optimized MongoDB cluster configurations, saving approximately \$12,000 annually while maintaining system performance and reliability.

- Led Kubernetes-based deployments and CI/CD pipeline improvements (Jenkins, GitHub Actions), incorporating Infrastructure as Code (IaC) practices to enhance scalability, resilience, and operational efficiency.
- Maintained real-time sync with Elastic indices.
- Created unit/behavior tests (Selenium, Chai, Mocha, Cypress) and integrated GitHub pipelines (JavaScript, Node.js).
- Configured Jenkins pipelines for automated test reporting.

## **iRide PK**

Jan 2025 – Sep 2025

*Flutter Developer (Contract/Part-Time)*

- Built the iRide ride-hailing app in Flutter, reaching 60,000+ users in Lahore, Pakistan within 2 months of launch.
- Used BLoC and Dart null safety to deliver robust, testable state management.
- Reduced MongoDB hosting costs from \$1,500/mo to \$300/mo (80% reduction) by implementing Redis caching, optimizing indices, and tuning queries.

## **Code For Pakistan (Volunteer)**

Mar 2021 – Present

*Web Engineer/Flutter Developer*

- Integrated a WhatsApp chatbot for the COVID-19 Telehealth Initiative, enabling identification and random assignment of high-risk patients to volunteer doctors (React, Node.js, PostgreSQL).
- Built a Flutter-based mobile app for lab evaluations (LAARC-standard surveys) paired with a Vue.js dashboard for performance reporting (Flutter, Vue.js, Node.js).
- Collaborated with WHO and NIH as a Solutions Engineer to architect and implement secure healthcare applications (HCAI app) using Flutter, Vue.js, Node.js, and MongoDB. Applied system design principles, microservices architecture, and performance tuning while documenting architecture and workflows to improve healthcare protocols and post-surgery infection tracking.
- Education Mentor, KP Women Program / WomenInTechPK (2023 – Present): conducted training sessions (e.g., Git and version control) for participants to enhance their collaborative development workflows.
- Facilitated the interview process in 2025 to hire mentors and candidates for the KP Women Program.
- Served as speaker and mentor for the KP Government Innovation Fellowship.

## **Jeddah Soft**

Sep 2018 – Dec 2018

*Mern Stack Developer (Remote Part Time)*

- *Doctor App* (MySQL, React, Node.js):
  - Built a React + Bootstrap dashboard for managing doctor appointments.
  - Implemented Highcharts for admin reporting.
  - Maintained a session-based shopping cart tied to user authentication.
- *ShopAPI (Open Source)*:
  - See the ShopAPI GitHub repo (Express/Node.js + WebSockets).
  - Uploaded and indexed files/images via Cloudinary.
  - Maintained session-based shopping cart logic.

*Internee*

- Developed a VR basketball game for HTC Vive.

**Skills**

- **Languages:** TypeScript, Java, Python, Go, Node.js, Dart, C++
- **Frameworks:** Angular, React, Vue, Ionic, Express, FastAPI, Flutter, Quarkus, Pandas, NumPy, Behave, Mocha, Chai, Cypress, Selenium, Protractor, Cucumber, Highcharts, Chart.js, Recharts, Bootstrap
- **Databases:** MongoDB, MySQL, MsSQL, PostgreSQL, ElasticSearch
- **AI & Automation:** n8n, RAG, OpenAI, IBM Watson (STT), Google Cloud Vision, GraphQL
- **Cloud & DevOps:**  AWS Certified (SNS/SQS), GCP (Compute Engine, Networking), Docker, Kubernetes, CI/CD, Jenkins, GitHub Actions, Apache, RabbitMQ, Kafka, Message Brokers, WebSockets, Cloudinary, Git (GitHub, GitLab)
- **Tools:** Jupyter, NetBeans, Eclipse, VS Code, HTC Vive
- **Architecture & Practices:** System Design, Microservices, API Gateway, Event-Driven Systems, Performance Optimization, Scalability, Cloud Native Architecture, System Architecture Patterns, Hybrid Mobile Architecture, Distributed Systems Architecture

**PROJECTS & RESEARCH**

Honeypot Analysis and Visualization Tool (FYP)

- Analyzed data from honeypots (Dionaea, Suricata, Kippo).
- Visualized data intensity on geographical maps.
- Centralized data in MongoDB for increased performance.

SWAM: Scalable Event-Driven Incident Management System

*Quarkus (Jakarta EE, JAX-RS, CDI, JPA), Angular, PostgreSQL, Kafka, Docker, Kubernetes*

*Personal project – see the full technical report for details*

- Designed a hexagonal (ports-and-adapters) architecture isolating the core domain behind a REST API built with Jakarta EE, JAX-RS, CDI, and JPA on Quarkus, with a lightweight CQRS read path.
- Implemented event-driven duplicate detection: the API publishes incident-creation events to a Kafka topic, consumed asynchronously by an independent analyzer service using fuzzy text-similarity matching, which reports back via a second topic without blocking the write path.
- Enforced role-based access control (Reporter, Department Admin, Super Admin) with department-scoped incident visibility and transactional write operations.
- Deployed via Docker Compose locally and Kubernetes in production, with the API service horizontally autoscaled (1–5 replicas, targeting 50% CPU) independently of the analyzer service.
- Verified the event-driven flow with Testcontainers-based integration tests and unit tests for domain logic.

Encrypted Document Vault with Sharing Control (Distributed Secure Platform)

*Go (Golang), Angular, RabbitMQ, MongoDB, Docker, WebSockets*

- Engineered a secure, distributed document vault platform for sharing sensitive files with full owner control, utilizing Go (Golang) for high-performance backend services and Angular/JavaScript for a responsive frontend.

- Designed a multi-instance distributed architecture to support horizontal scaling, ensuring system-wide state consistency (e.g., document deletion/access revocation) by using RabbitMQ (AMQP) for reliable event dispatching across all instances.
- Implemented a real-time “Eye Tracking System” via WebSockets to instantly notify document owners of current viewers, tracking access by IP address and browser fingerprint for traceability.
- Enforced comprehensive security, including AES-256 file encryption with user-specific keys, configurable time-limited/burn-once links, and instant system-wide access revocation.
- Integrated anti-capture measures including watermarking and client-side hardening to restrict copying, downloading, and developer tool access within the document view.
- Utilized Docker and Docker Compose for containerization and deployment, structuring the backend with a clean repository pattern using Go generics for testability and separation of concerns.

## **Achievements & Leadership**

### **Università degli Studi di Firenze**

Jul 2025 – Present

#### *Student Representative*

- Member of the Commissione Paritetica Docenti-Studenti CdS, serving as a bridge between the student body and faculty.
- Contribute to the evaluation and improvement of teaching quality and academic services within the Software, Science and Technology program.

### **WomenInTechPK**

Feb 2023 – Present

#### *Education Mentor*

- Lead hands-on sessions and mentoring programs to help participants gain real-world experience with modern tech tools, frameworks, and workflows used in industry and civic tech.

### **TEN**

2017 – 2018

#### *Alumni Relations Manager*

- Managed alumni network relations and engagement.